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LKS Faculty of Medicine
Department of Orthopaedics
& Traumatology
香港大學矯形及創傷外科學系



Knee Osteoarthritis

Part A



Professor Lewis PK Chan
Clinical Associate Professor

Objectives

- Epidemiology & clinical importance of knee osteoarthritis (KOA)
- Pathophysiology & risk factors for developing KOA
- Clinical presentation & workup of KOA
- Non-pharmacological & pharmacological treatment options
- Surgical interventions for managing KOA

Case studies



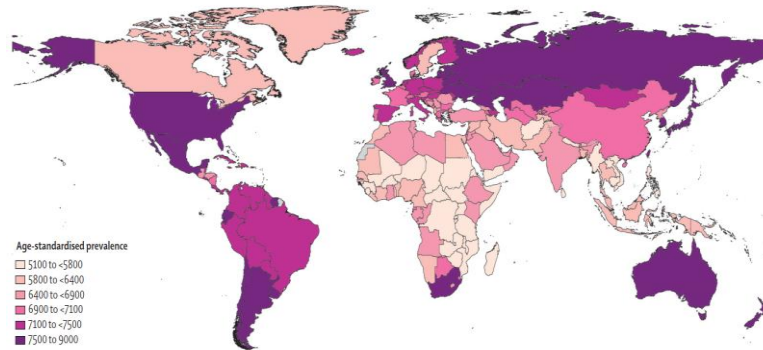


1. Introduction

Introduction

- The Global Burden of Disease (GBD) study

- **Knee Osteoarthritis (KOA)** is a major cause of adult disability
- estimate - almost 1 billion individuals in the year 2050



Lancet Rheumatol 2023

- Hong Kong

- By 2050 (forecasted by the World Health Organization)
 - 40% of the population in Hong Kong will be 65 or above
 - rank fifth in the world



Population ageing trend of Hong Kong, Office of the Government Economist, HKSAR, 2023

1. OA is a Serious Disease ?



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- Increased risk of cardiovascular/respiratory disorder
- Increased risks of Psychological Disorder
- Increased risks of sleep disturbance
- Increased risk of Mortality

T. Kasai et al. Journal of Orthopaedic Science 22 (2017) 1126-1131

The impact of musculoskeletal diseases on mortality-comparison with internal diseases: A 15-year longitudinal study



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- 601 Japanese subjects (Mean Age 67.8+/-5.3)

- FU 2000 – 2014 (15 years)

- Significant **Higher Risk of Mortality (HR**

- **Knee OA (1.97)**

- Osteoporosis (1.61)

- DM (1.97)

- HT (1.39)

- **Common Cause of Death**

- **Cardiovascular disease**

OA Can **KILL** you !!



Multivariable analysis (Total subjects).

	Hazard ratio	95% CI	P-value
Smoking	1.706	1.141–2.552	0.009
Alcohol drinking	0.880	0.598–1.294	0.516
Exercise	1.194	0.814–1.753	0.364
Osteoarthritis	1.972	1.356–2.867	<0.001
Osteoporosis	1.614	1.126–2.313	0.009
Diabetes	1.965	1.146–3.368	0.014
Hypertension	1.391	0.977–1.982	0.067
Hyperlipidemia	1.063	0.724–1.560	0.755

95% CI: 95% confidence interval. Adjusted for age, sex, and BMI.

T. Kasai et al. Journal of Orthopaedic Science 22 (2017) 1126-1131

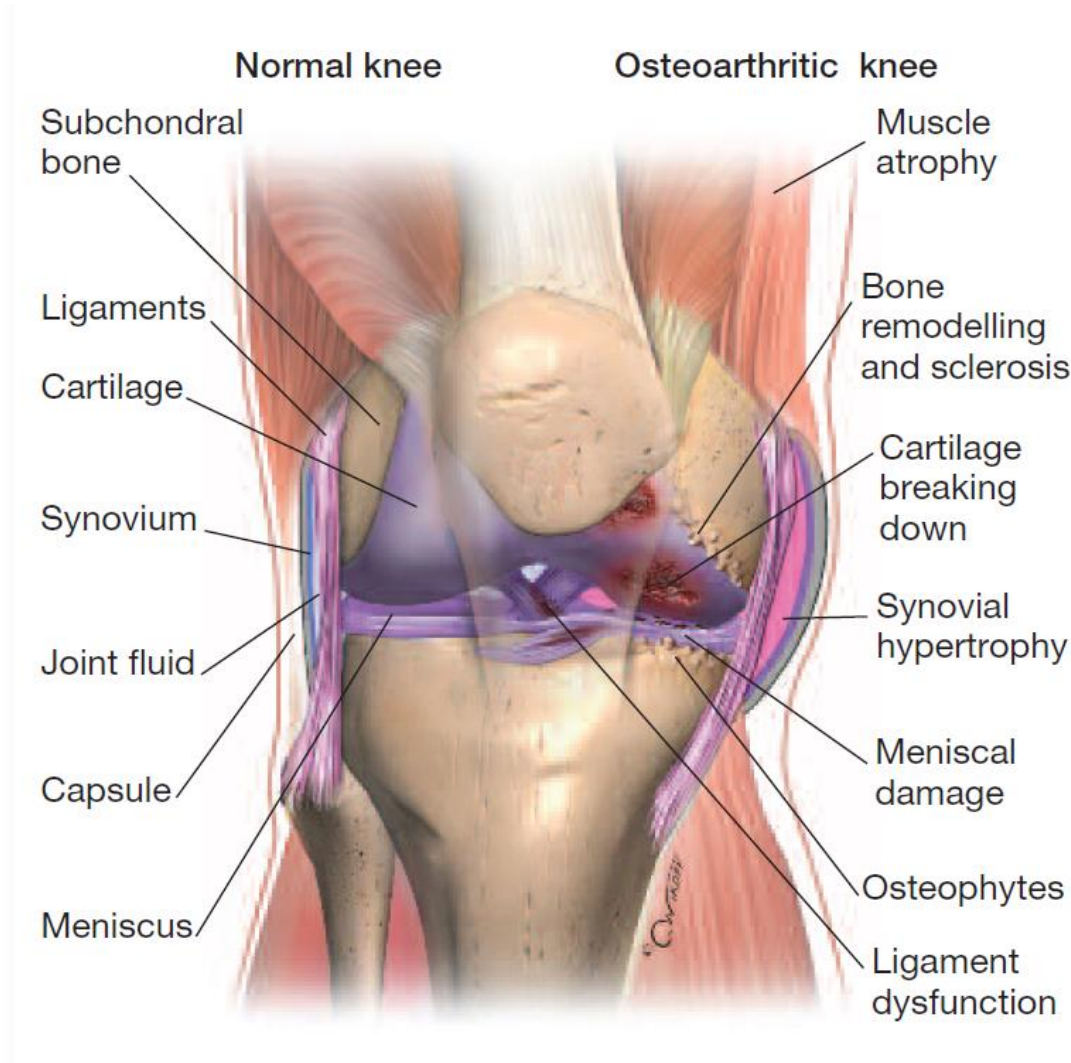
2. Is OA just a cartilage disease of Wear & Tear ?



- Wear and Tear Mindset result in
 - Inactivity



OA is a Whole Joint Disease



Importance of Exercise

Muscle

- Exercise to prevent Atrophy

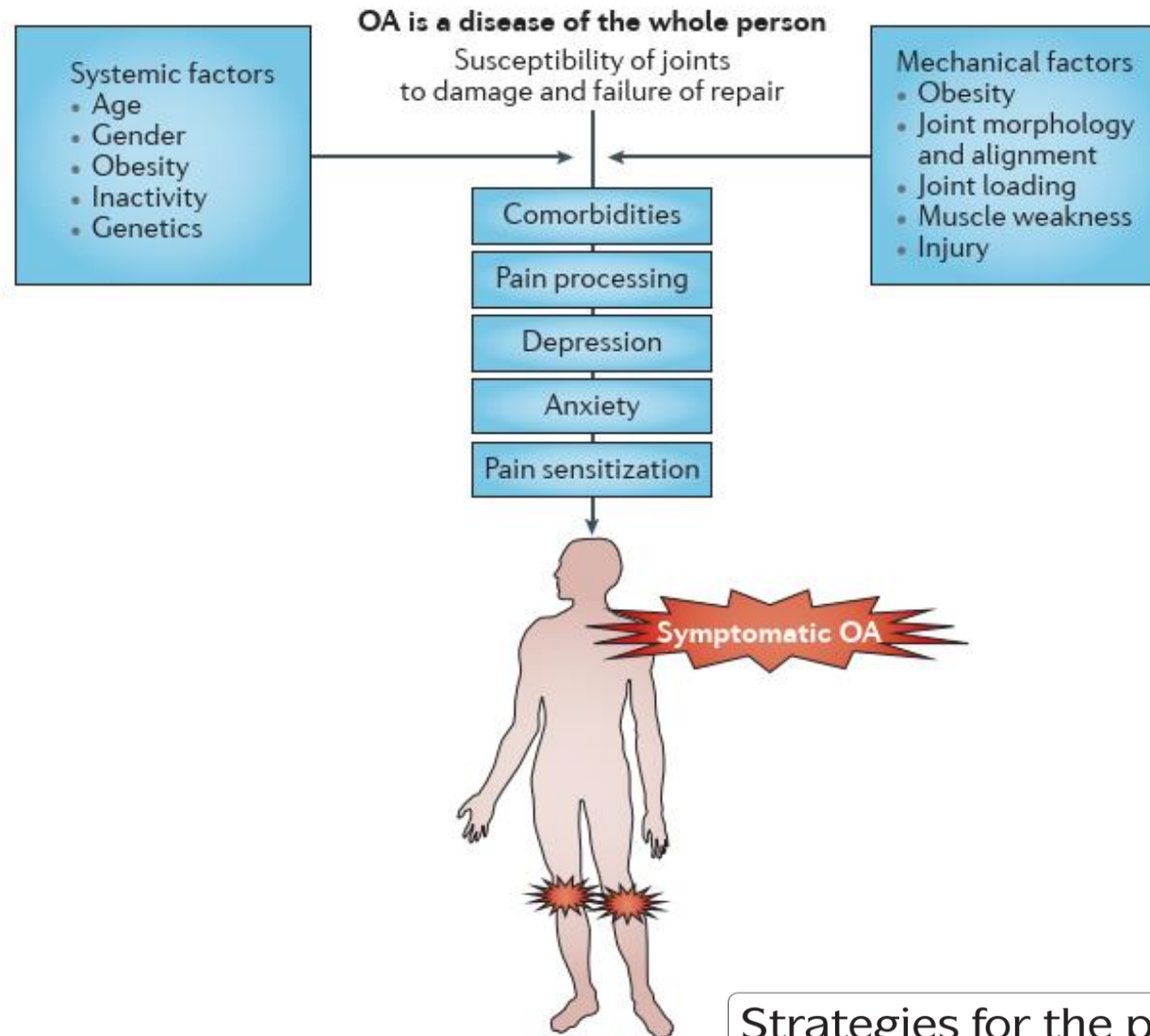
Cartilage Health

- Repeated Cyclical Loading

OA is a Whole Person Disease



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Traditional Patient Journey in OA knee

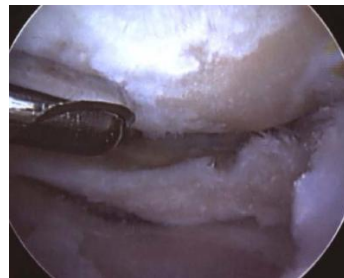


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Waiting Time for New Case Booking at Orthopaedics & Traumatology Specialist Out-patient Clinics

	New case bookings ₍₂₎		Waiting Time							
	Number	Percentage		Hong Kong East	Hong Kong West	Kowloon Central	Kowloon East	Kowloon West	New Territories East	New Territories West
Stable Case	76,228	72%	Median	51 weeks	20 weeks	36 weeks	57 weeks	50 weeks	59 weeks	36 weeks
			Longest ₍₁₎	87 weeks	89 weeks	98 weeks	75 weeks	92 weeks	90 weeks	58 weeks

Hospital Authority Statistics : In the past 12 months (01 April 2023 - 31 March 2024)



Patients are NOT Unsatisfied: High Unmet Medical Need



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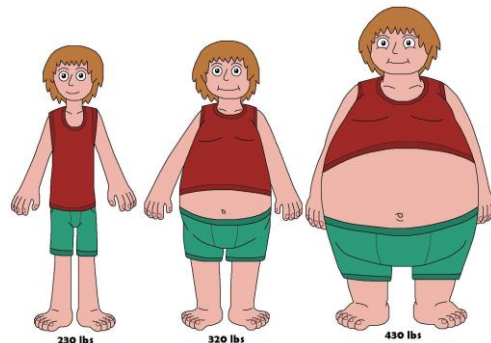
Today's treatment paradigm is trapping patients in a vicious cycle of OA knee pain

Peplemetrics Patient Segmentation Research, 2007



81%

Knee OA patients
highly unsatisfied with
current treatment



– Better Management of Osteoarthritis initiative in Sweden

2004

Clinical Guidelines for Managing Lower-limb Osteoarthritis in Hong Kong Primary Care Setting

Guideline Development Panel

Professor Albert Lee, Department of Community and Family Medicine, CUHK

Dr Clement Tsang, Department of Community and Family Medicine, CUHK

Professor William Wong, Department of Community and Family Medicine, CUHK

Professor Samuel Wong, Department of Community and Family Medicine, CUHK

Guideline Review Panel

Professor K.M. Chan, Chair Professor, Department of Orthopaedics and Traumatology, CUHK

Dr Timothy C.Y. Kwok, Associate Professor, Division of Geriatrics, Department of Medicine and Therapeutics, CUHK

Professor Kenneth K.C. Lee, Professor, School of Pharmacy, CUHK

Dr K.Y. Lee, Chief Hospital Manager and Specialist in Rheumatology, Union Hospital

Dr Arran Leung, part-time Assistant Professor (Physiotherapist), Centre for Health Education and Health Promotion, CUHK



Non-surgical treatment of knee osteoarthritis

HS Kan, PK Chan *, KY Chiu, CH Yan, SS Yeung, YL Ng, KW Shiu, Tegan Ho

ABSTRACT

Knee osteoarthritis is one of the most common degenerative diseases causing disability in elderly patients. Osteoarthritis is an increasing problem for ageing populations, such as that in Hong Kong. It is important for guidelines to be kept up to date with the best evidence-based osteoarthritis management practices available. The aim of this study was to review the current literature and international guidelines on non-surgical treatments for knee osteoarthritis and compared these with the current guidelines in Hong Kong, which were proposed in 2005. Internationally, exercise programmes for non-surgical management of osteoarthritis have been proven effective, and a pilot programme in Hong Kong for comprehensive non-surgical knee osteoarthritis management has been successful. Long-term studies on the effectiveness of such exercise programmes are required, to inform future changes to guidelines on osteoarthritis management.

Hong Kong Med J 2019;25:Epub
<https://doi.org/10.12809/hkmj187600>

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Hong Kong Med J 2019 Apr;25(2):127–33

2019

Feasible non-surgical options for management of knee osteoarthritis during the COVID-19 pandemic and beyond

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Hong Kong Med J 2024;30:56–61
<https://doi.org/10.12809/hkmj2210209>

Hong Kong Med J 2024 ;30:56–61

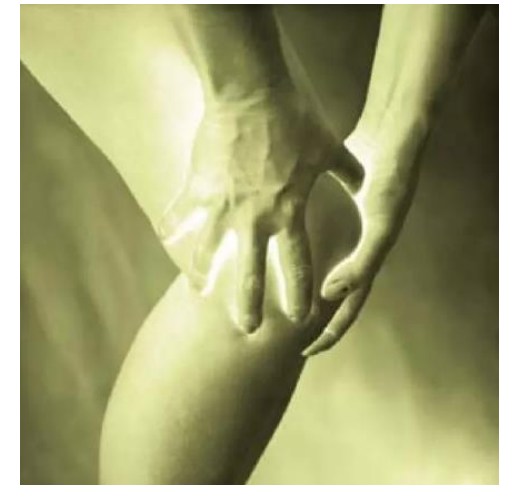
2. Clinical Assessment of Knee Osteoarthritis

- IHD/AF on apixaban, Hx of gastritis
- Complain of Knee pain
- How do you assess?



Chief Complaints

- Pain
- Swelling
- Stiffness
- Mechanical Disorders (locking, giving Way, click)
- Limp
- Deformity



Knee Pain : Location

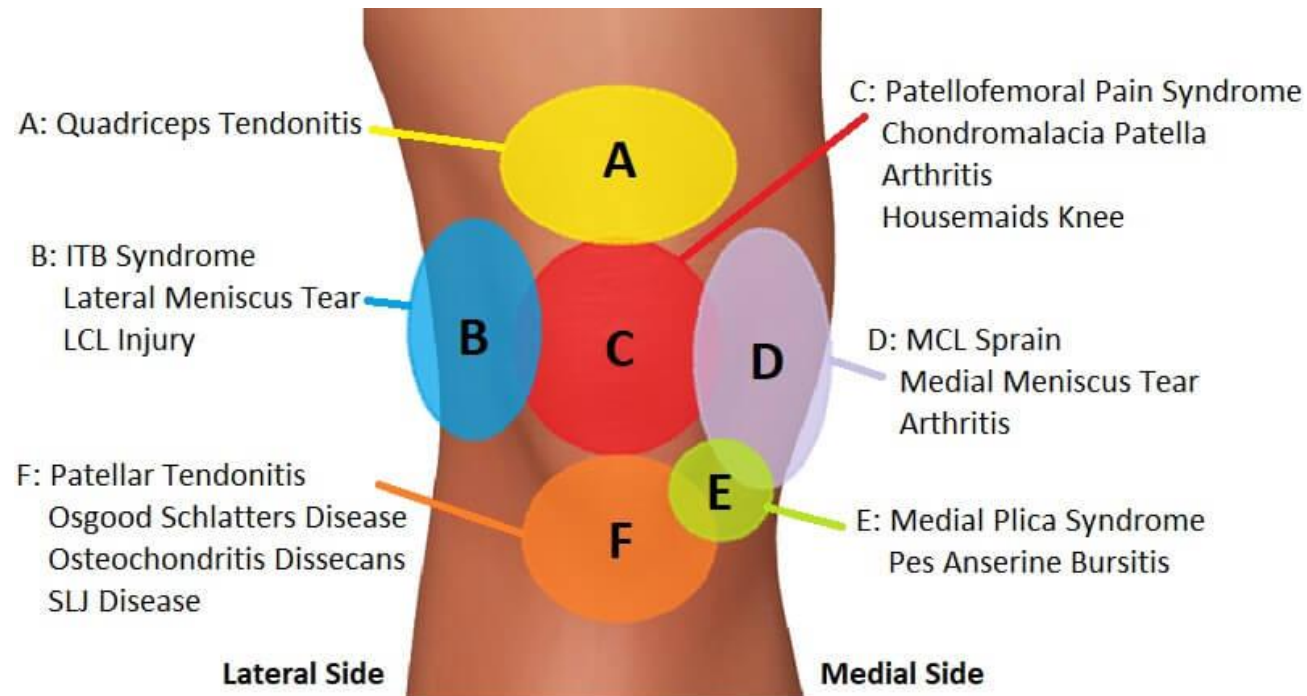


TABLE 2

Differential Diagnosis of Knee Pain by Anatomic Site

Anterior knee pain

Patellar subluxation or dislocation
Tibial apophysitis (Osgood-Schlatter lesion)
Jumper's knee (patellar tendonitis)
Patellofemoral pain syndrome (chondromalacia patellae)

Medial knee pain

Medial collateral ligament sprain
Medial meniscal tear
Pes anserine bursitis
Medial plica syndrome

Lateral knee pain

Lateral collateral ligament sprain
Lateral meniscal tear
Iliotibial band tendonitis

Posterior knee pain

Popliteal cyst (Baker's cyst)
Posterior cruciate ligament injury

Importance of Trauma History: Knee pain after Injury



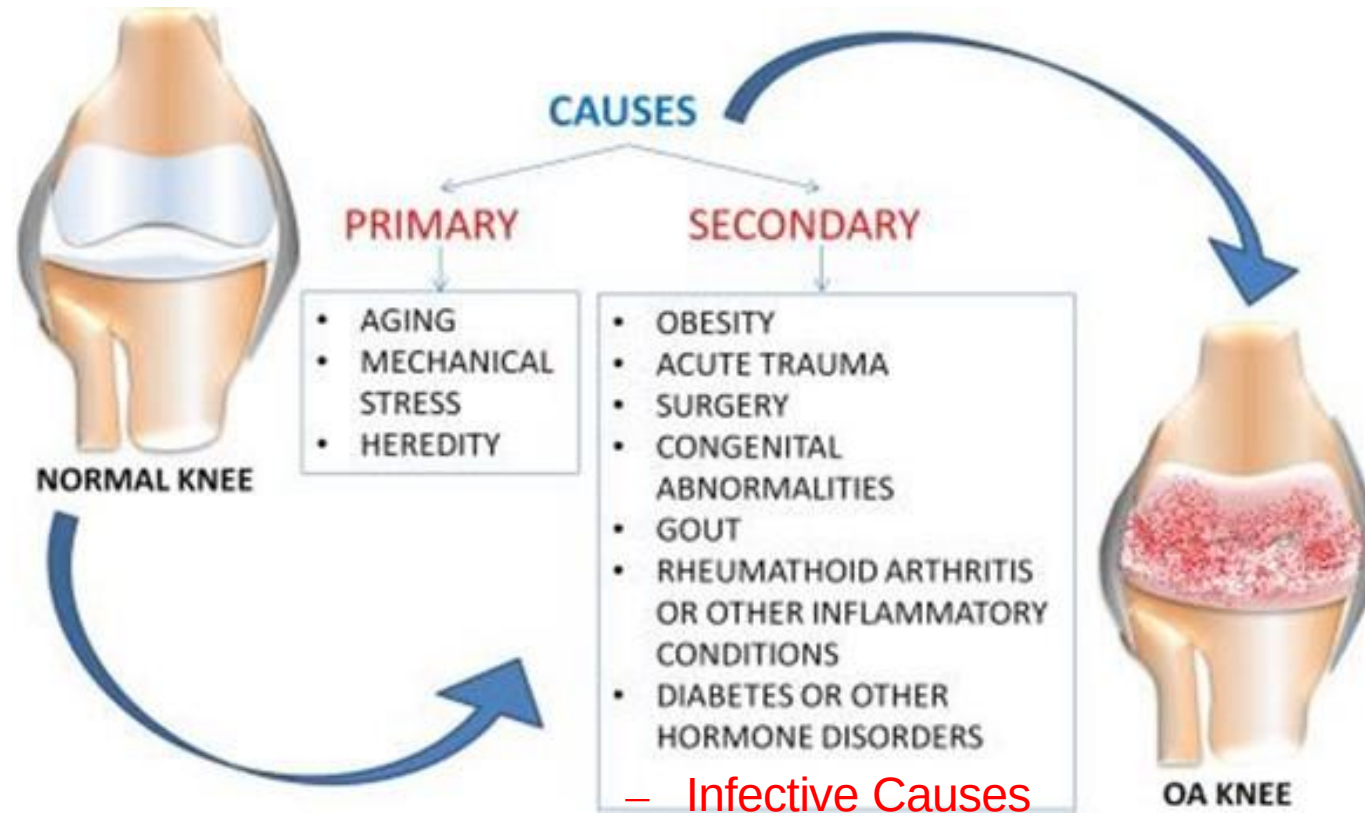
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John B. Imboden, David B. Hellmann, John H. Stone+

Table 12–2. Essential Features of Acute Knee Injuries

	Joint Swelling	Characteristics
ACL/PCL tears	Immediate	Instability
Meniscus	Delay (first 24 hour)	Pain with twisting
Patella dislocation	Immediate	Pain with palpation of medial facet, lateral femoral condyle
Fractures	Immediate	Unable to bear weight

Causes



Who is at risk of Osteoarthritis ?



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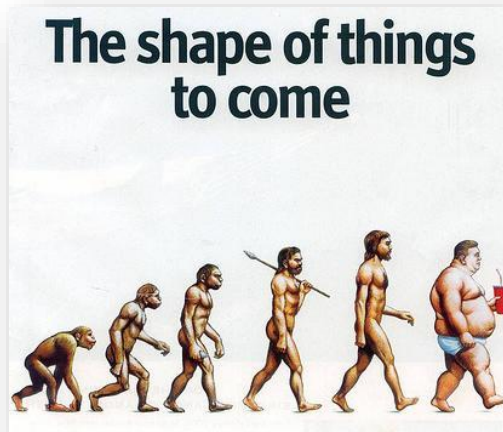
- NBA superstar Shaquille O'Neal has had osteoarthritis for over a decade.
- Osteoarthritis is more common in people
 - at the extremes of body weight
 - often takes a toll on weight-bearing joints like the knees, hip.
 - THA done in 2023



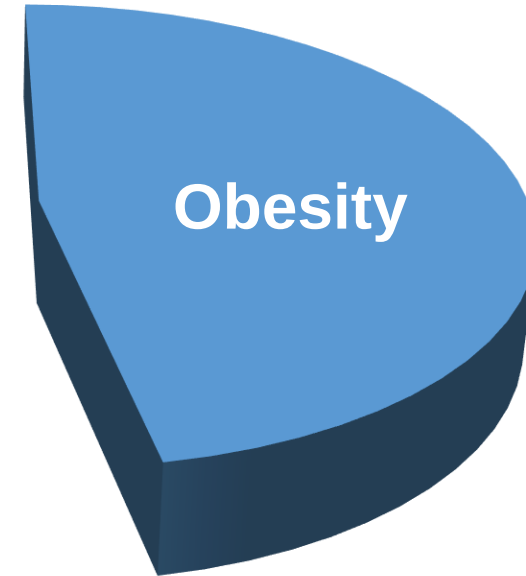
Risk Factors for Knee OA



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Proudly supported by



Arthritis Rheum. 1998, Aug;41(8):1343-55.
Osteoarthritis Cartilage. 2009; Sep 2.

Orthopaedic Physical Exam



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– Look

– Feel

– Move

Knee Deformity on Standing



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20.1 Examination standing (a) Look at the general shape and posture, first from in front and then from behind. Normally the knees are in slight valgus. (b) Varus deformity (osteoarthritis). (c) Valgus deformity (rheumatoid arthritis).

Knee Examination - Move



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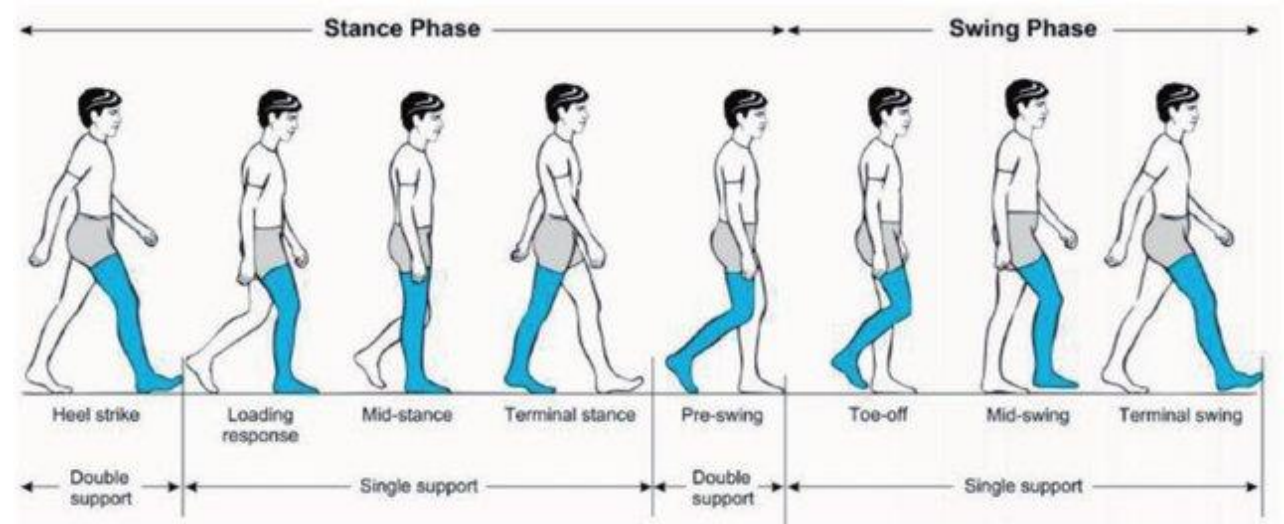
- Fixed flexion deformity
 - limited knee extension range both actively and passively
- Extension Lag
 - lack of active knee extension with full passive knee extension
- Recurvatum
 - knee hyper-extension greater than 5 degrees



Gait Exam : 77 Year old Female



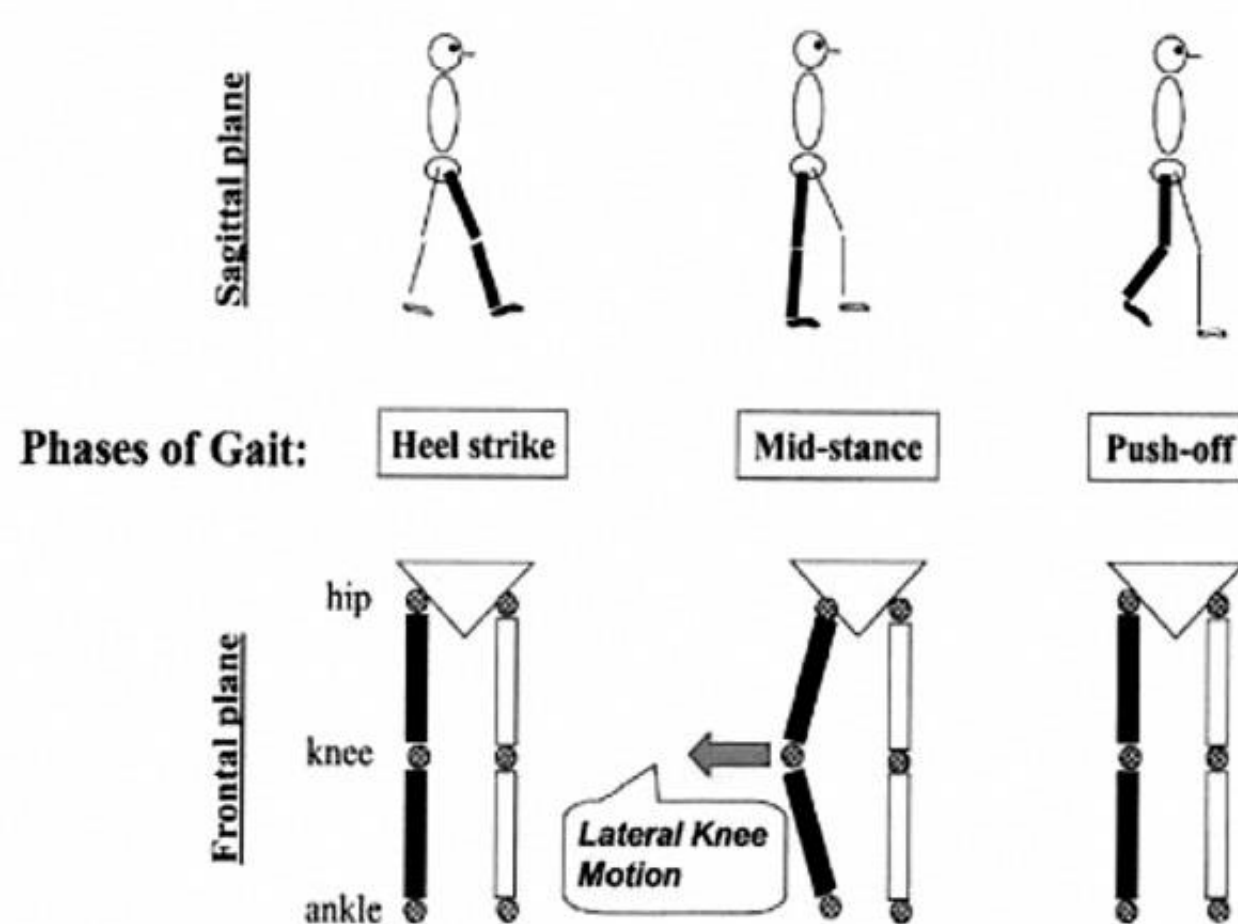
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Varus thrust



worsening of varus alignment while the limb is bearing weight, with return to less varus as weight is transferred off the limb.

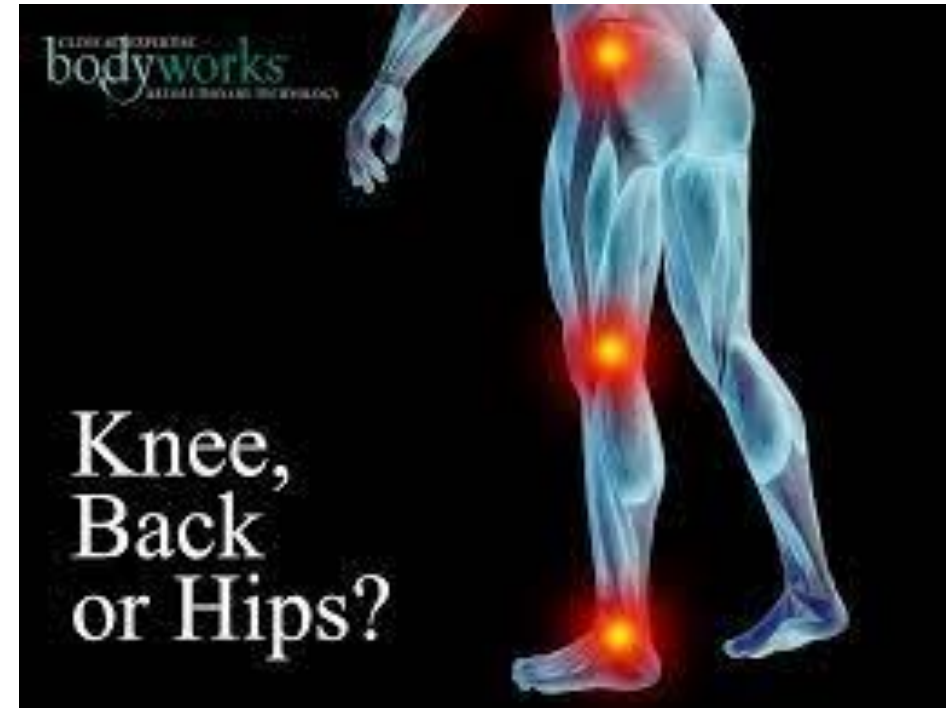


Every patient with knee pain should have a careful examination of the hip



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- Pain can be referred to the knee from other sites, most notably the ipsilateral hip.





3. Workup of Knee Osteoarthritis

Any Investigations ?



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Non-Weight-bearing XR



Weight-bearing XR



Standard XR Knee



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- Wt bearing AP, lateral
- Skyline



OA : Radiographic features



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- The hallmarks of knee osteoarthritis
 - joint space narrowing
 - subchondral sclerosis
 - marginal osteophytes
 - subchondral cysts

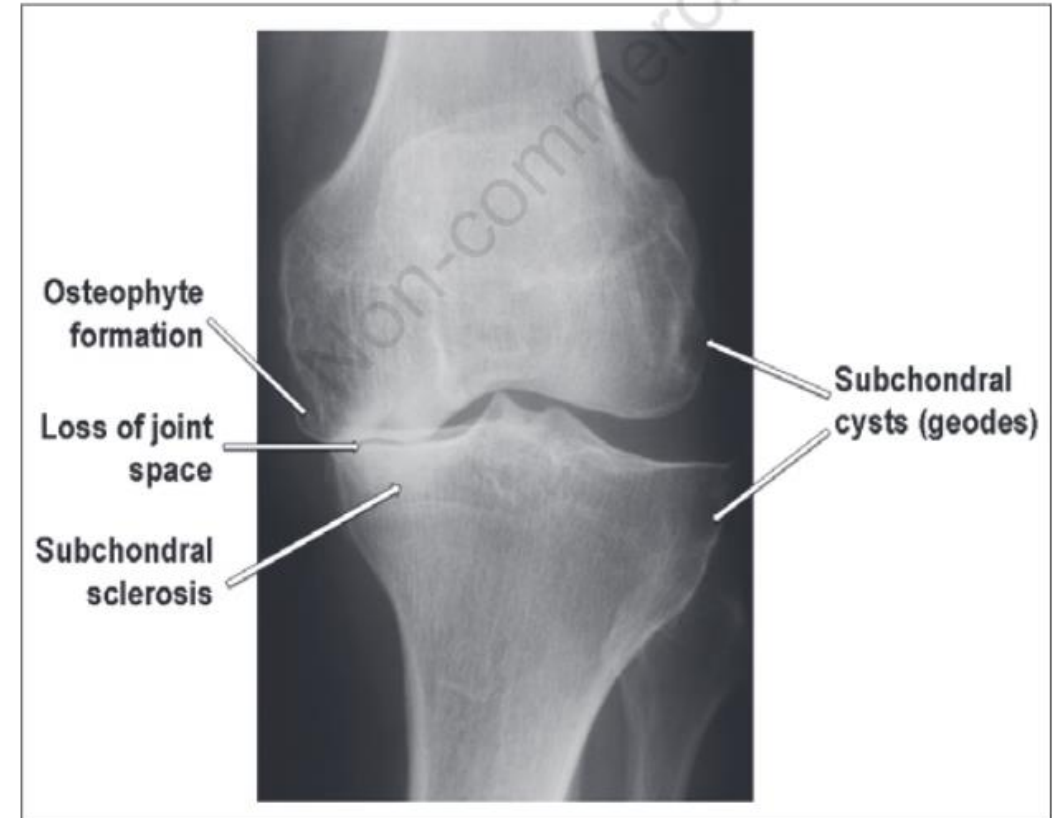
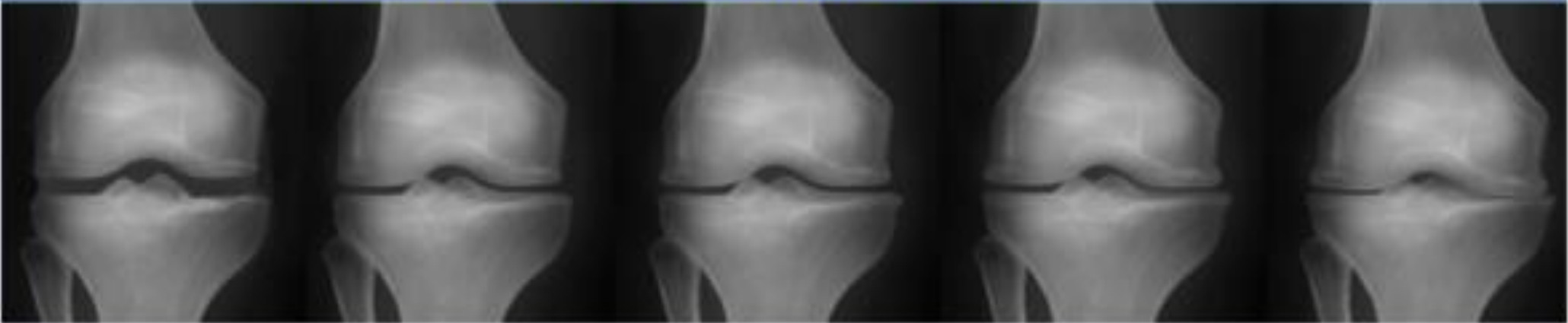


Figure 3 - Anteroposterior radiograph of the knee of a patient with osteoarthritis. Note the varus malalignment (resulting in *bow-legs*) due to loss of cartilage and joint space in the medial compartment, subchondral cysts and osteophyte formation.

Grading of Severity of Knee OA: Kellgren and Lawrence system



Kellgren–Lawrence grading scale					
X-Ray					
OA Grade	Grade 0 (Normal)	Grade 1 (Doubtful)	Grade 2 (Mild)	Grade 3 (Moderate)	Grade 4 (Severe)
JSN	No radiographic features of OA are present	Doubtful	Possible	Definite	Marked
Osteophytes		Possible	Definite	Multiple	Large

4. Non-Pharmacological Mx of Knee Osteoarthritis

4. What is The Core Non-pharmacological Mx of Knee OA ?



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Surgery

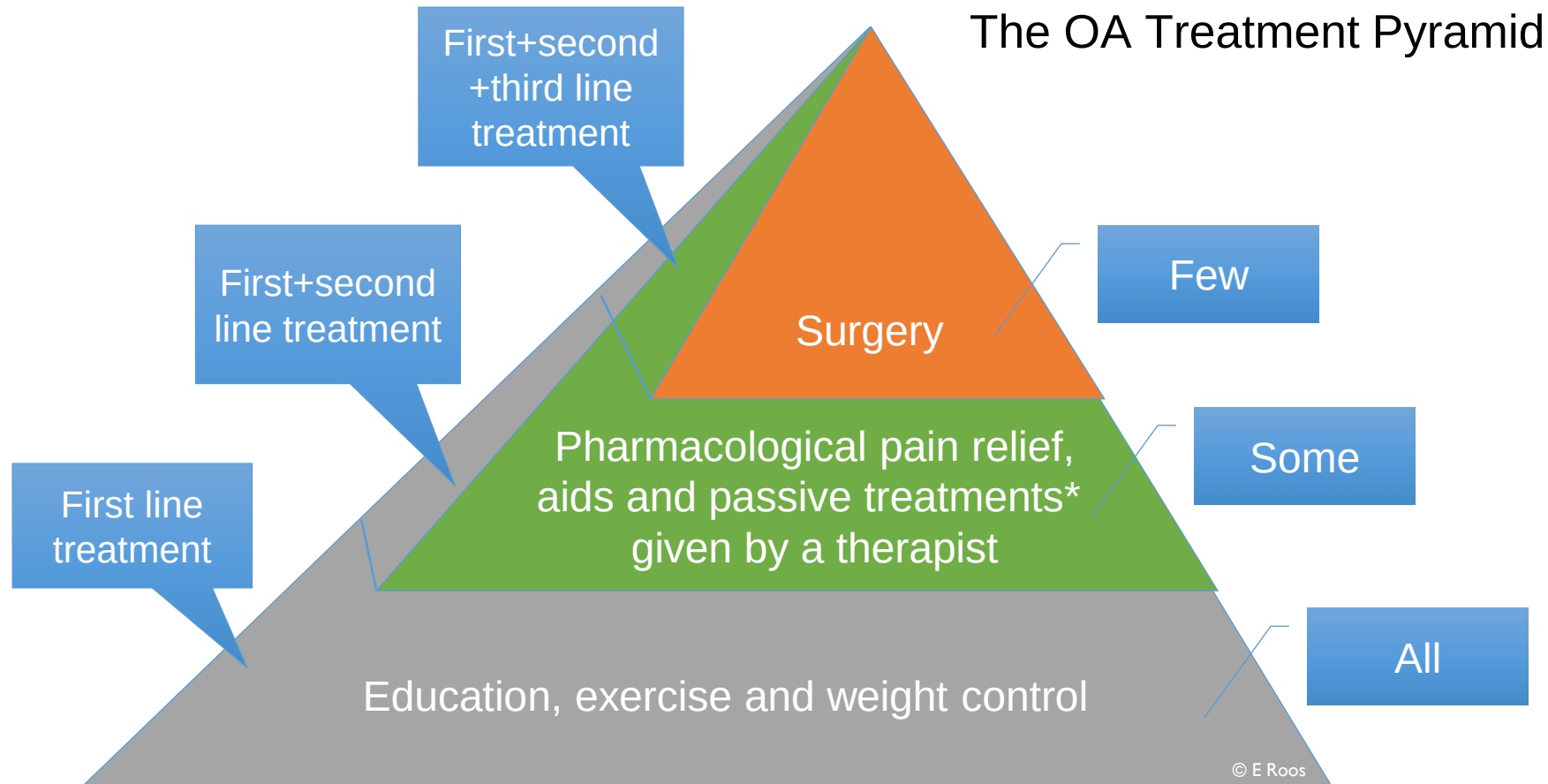
Pharmacological pain relief,
aids and passive treatments*
given by a therapist

Education, exercise and weight control

© E Roos

Roos & Juhl. Osteoarthritis and Cartilage 2012

4. The Core Mx of OA : Education, Exercise and Weight Control



Passive modality in Physiotherapy DON'T work in long run



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- Public perception of Physiotherapy



- Evidenced-based PT
 - Active Exercise



Physiotherapy



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大腿的股四頭肌及腿後肌是鍛練的重點

Exercise Need to have sufficient intensity, frequency & duration



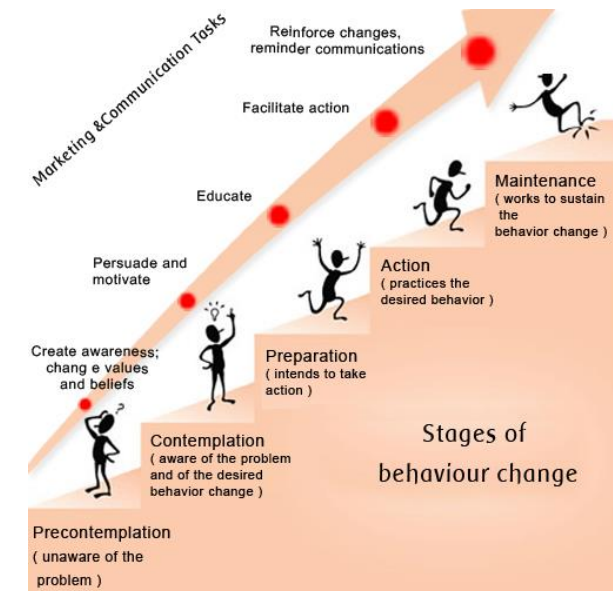
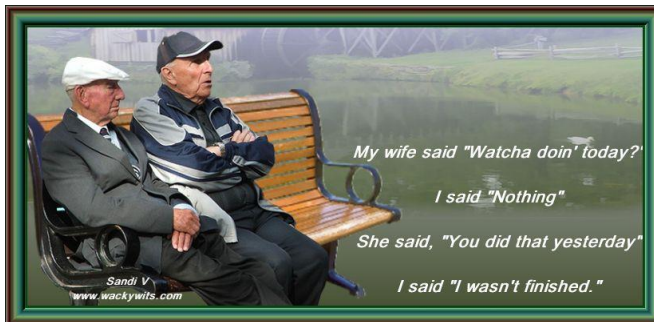
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- Systematic Review and meta-analysis, 48 trials:
- **Sufficient dose**
 - Focus on improving **aerobic capacity, quadriceps muscle strength or lower extremity performance**
 - **Supervised, Two or three times weekly for 6 weeks**
- **To enhance compliance in exercise**

Juhl et al. Arthritis & Rheumatology 2014



- Behavioral Change Model
Individual Goal Setting



MMRC Jockey Club Specialist OPD Clinic



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- Approval of donation by The Hong Kong Jockey Club Charities Trust (HKJCCT)
- Approved grant: \$20.27Million

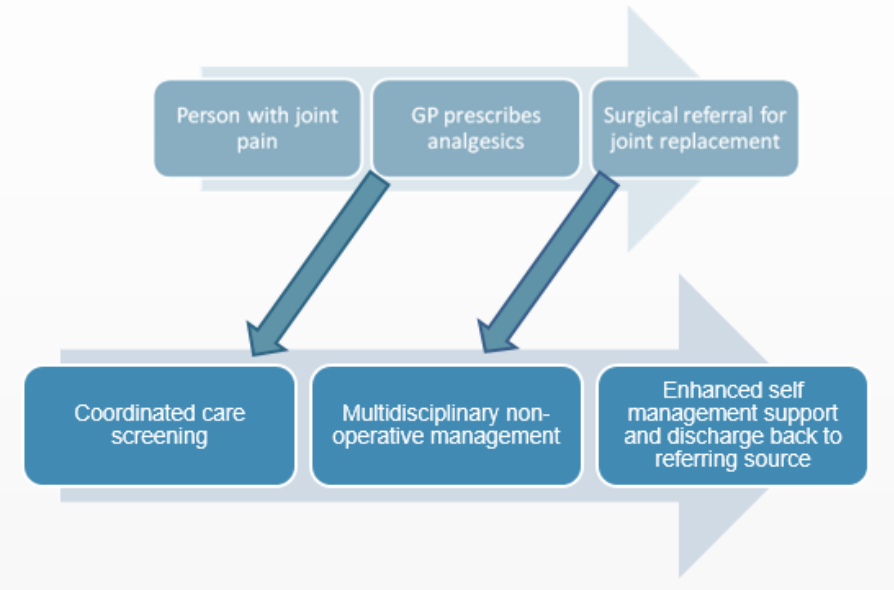


COME Multidisciplinary Team



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- O&T Surgeon
- Nurse
- Physiotherapist
- Occupational Therapist
- P&O
- Dietician
- Psychologist
- Pharmacist



麥理浩復康院 退化性關節管理小組

本港首間 全方位 關節退化 非手術治療中心

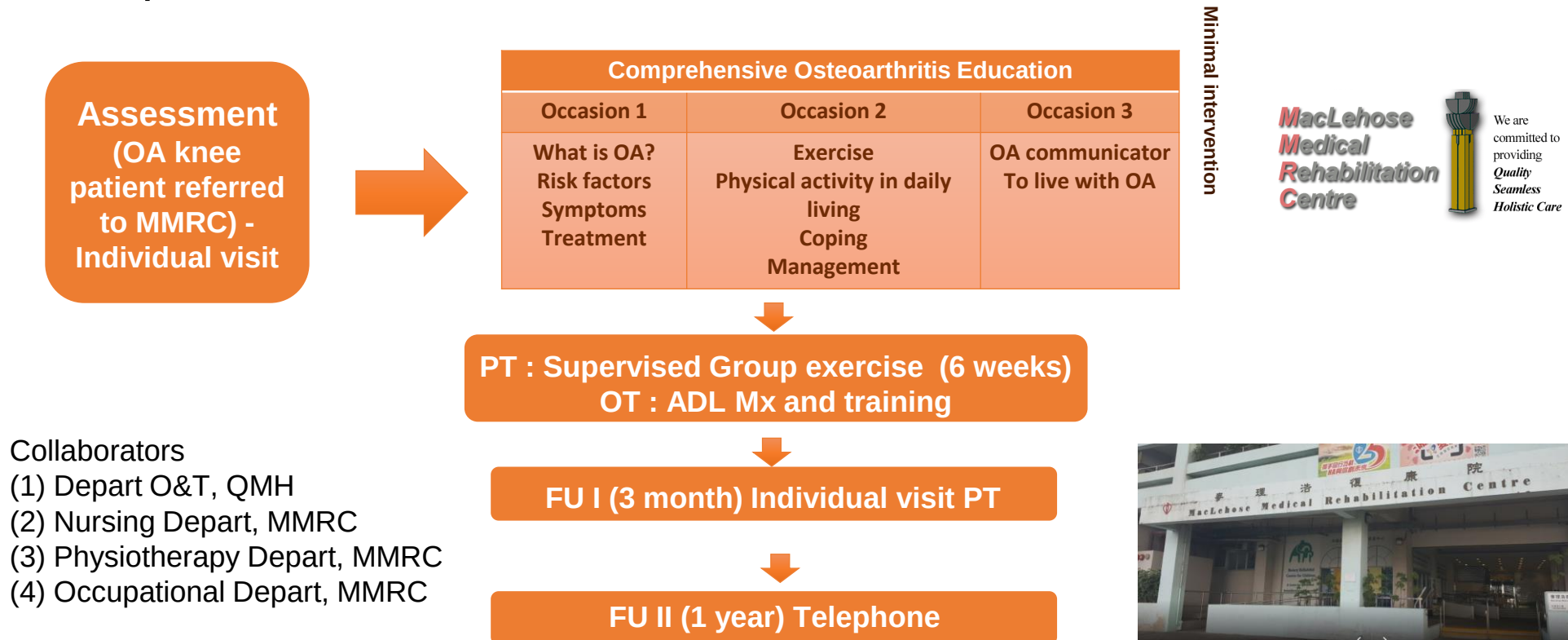
COME (Comprehensive Osteoarthritis Management) in MMRC



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- Aim

To provide comprehensive Non-surgical Management care for patient with osteoarthritis



Result of COME Programme at MMRC



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Non-surgical treatment of knee osteoarthritis

HS Kan, PK Chan *, KY Chiu, CH Yan, SS Yeung, YL Ng, KW Shiu, Tegan Ho

TABLE 3. Patients receiving Comprehensive Osteoarthritis Management (COME) showed consecutive significant improvements over each assessment time point (n=55)*³¹

	Data			
Age (years)	63.3 ± 7.4			
Gender				
Male	13			
Female	42			
	Baseline	6 Weeks	3 Months	12 Months
PSEQ score	42 ± 11.2	46.7 ± 10†		
FACIT score	34.4 ± 8.5	38.4 ± 7.1‡		
One-minute chair test repetitions	26.8 ± 11.3	35.5 ± 12.2‡	38.1 ± 11.1‡	
Right quadriceps strength (kg)	25.6 ± 8.9	29.9 ± 9.9‡	31.1 ± 11.0‡	
Left quadriceps strength (kg)	24.8 ± 9.2	28.0 ± 9.2‡	30.5 ± 10.4‡	
PSFS score	3.5 ± 2.3	6.7 ± 1.8‡	6.8 ± 1.8‡	6.7 ± 1.9‡
SEE-C score	47.3 ± 14.0	64.9 ± 11.1‡	65.2 ± 12.4‡	62.0 ± 17.6‡
Weekly time spent for exercise (mins)	58.4 ± 44.9	81.8 ± 50.5‡	88.4 ± 33.8‡	107.5 ± 26.3‡

Successful Result of COME Programme



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– F/64

- Minimal knee pain (VAS 1-2)
- Walk unaided > 1 hr
- Exercise everyday
- NOT want TKA



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